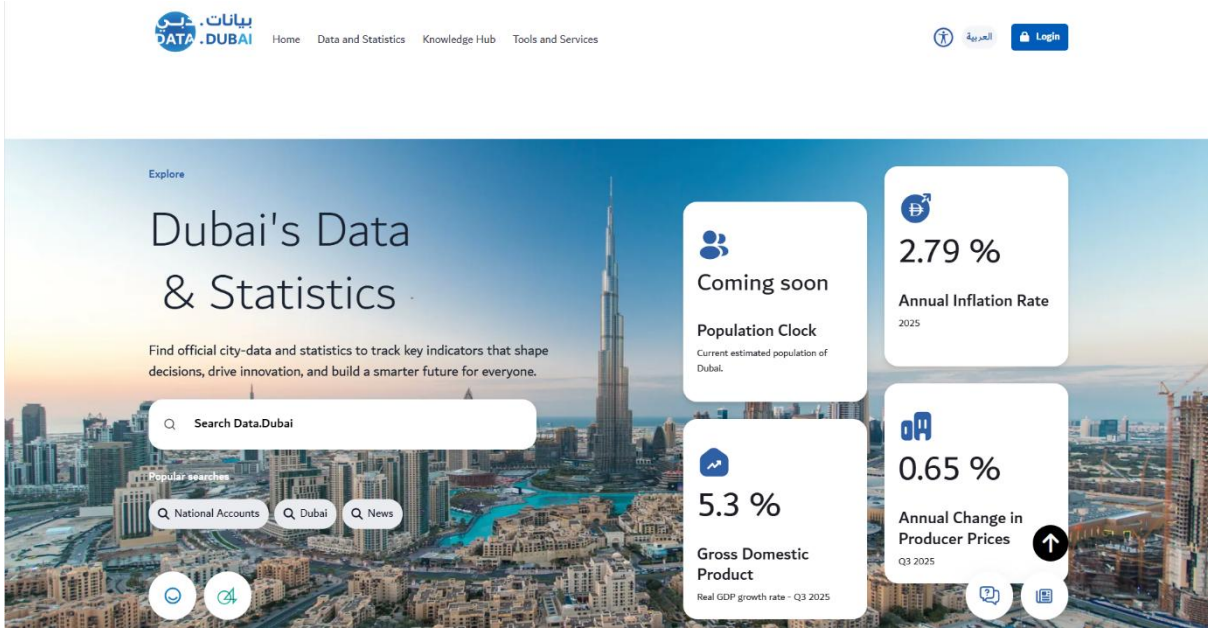

API Usage User Guide

User Guide for data.dubai API Invocation

Digital Dubai Authority
data.dubai

Introduction

This user guide provides a clear and structured overview of how to access and invoke the data.dubai APIs. It is designed to help users understand the key steps involved in connecting to the APIs, making requests, and interpreting responses effectively.

The document outlines the essential prerequisites, authentication mechanisms, and request/response formats required for successful API integration. It also includes practical guidance to ensure a smooth and efficient onboarding experience for users.

By following this guide, users will be able to confidently interact with data.dubai APIs and integrate them into their applications or workflows with ease.

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1. Pre-Requisites

Before accessing data.dubai APIs, ensure that the following requirements are met:

1. **User registration**
   Users must be registered on data.dubai platform with a valid email address to gain access to available services.

2. **API access request**
   This can be achieved by requesting API access for the desired resource on the portal.

3. **Client Credentials for Authentication**
   Once the API access request is completed dedicated client credentials (such as client_id and client_secret) will be provided. These credentials can be used to authenticate API requests.

Ensuring these prerequisites are in place will enable a seamless and secure interaction with the data.dubai APIs.

2. Accessing APIs

The data.dubai APIs use an **OAuth 2.0**-based authentication mechanism to ensure secure access. To successfully invoke the APIs, follow the steps outlined below:

1. **Generate Bearer Token**
   Begin by invoking the Token API using the provided client credentials. Upon successful authentication, a Bearer Token will be issued. This token is required for all subsequent API calls.

2. **Validate Connectivity (Health-Check API)**
   It is recommended to invoke the Health Check API, at least once when using the APIs for the first time, using the generated Bearer Token. This step helps confirm that your requests are correctly reaching the application and that authentication is functioning as expected.

3. **Invoke Data API**
   Once the token is validated, you can proceed to invoke the Data API using the same Bearer Token to retrieve the required data from data.dubai.

Detailed information regarding required headers, parameters, and request formats for each API is provided in the following sections to assist with proper implementation.

3. Generate bearer token

The Token API is used to generate a **Bearer Token**, which is required to authenticate and authorize all subsequent requests to the data.dubai APIs.
To obtain the token, the client must provide valid authentication credentials.

3.1 API Details:

Key Details

- **Purpose:** Generate an access token for API authentication
- **Authentication Type:** OAuth 2.0 (Client Credentials)
- **Token Validity:** 3600 seconds (1 hour)

Request Requirements

- **Method:** POST
- **Headers:**
  - Content-Type: application/json
  - x-DDA-SecurityApplicationIdentifier: <provided by integration team>

- **Query Parameters:**
- **Body Parameters:**
  - **grant_type**: Should be set to "**client_credentials**"
  - **client_id**: Provided client identifier
  - **client_secret**: Provided client secret

Response

On successful authentication, the API returns a JSON response containing:

- **access_token**: The Bearer Token to be used in subsequent API calls
- **token_type**: "Bearer"
- **expires_in**: Token validity duration in seconds

This access token must be included in the **Authorization header** of all subsequent API requests as follows:

Authorization: Bearer <access_token>

Note: Upon testing the Non-production API, kindly reach out via data Portal website under "contact us" form with your Application ID for production credentials. Test and Production Credentials are not same.

3.2 How to Use?

Sample Request

```
curl --location 'https://stg-apis.data.dubai/secure/ssis/dubaiai/gatewaytoken/1.0.0/getAccessToken' \
```

```
--header 'Content-Type: application/json' \
--header 'x-DDA-SecurityApplicationIdentifier: <Identifier key provided by integration team>\
--data '{
   "grant_type": "client_credentials",
   "client_id": "my-client-id",
   "client_secret": "my-client-secret"
}'
```

```
curl --location 'https://apis.data.dubai/secure/ssis/dubaiai/gatewaytoken/1.0.0/getAccessToken' \
--header 'Content-Type: application/json' \
--header 'x-DDA-SecurityApplicationIdentifier: <Identifier key provided by integration team>\
--data '{
   "grant_type": "client_credentials",
   "client_id": "my-client-id",
   "client_secret": "my-client-secret"
}'
```

Sample Response

```
{
   "access_token": "xxxxx",
   "token_type": "Bearer",
   "expires_in": 3600,
   "scope": "authz"
}
```

Sample Request on Postman

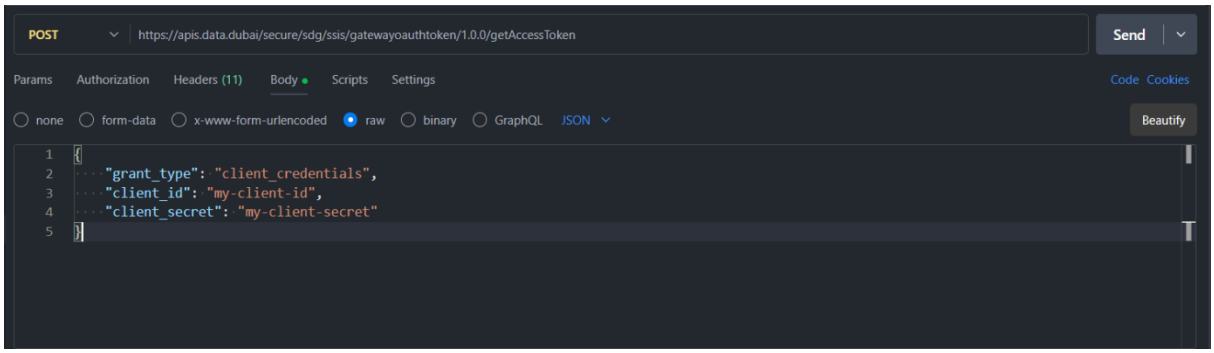


Figure: Token API Postman Request

Ensure that the token is refreshed once it expires to maintain uninterrupted access to the APIs.

4. Validate connectivity (Health-Check API)

The Health Check API is used to verify the availability and accessibility of the data.dubai API services. It serves as a quick validation step to ensure that authentication is successful and that requests are correctly reaching the application.

4.1 API Details:

Key Details

- **Purpose:** Validate API connectivity and authentication
- **Authentication:** Requires a valid Bearer Token
- **Method:** GET

Request Requirements

- **Headers:**
  - Authorization: Bearer <access_token>

Response

On successful invocation, the API returns a response indicating that the service is up and running. This typically includes a status message confirming the health of the application.
If the request fails, it may indicate issues such as:

- Invalid or expired token
- Incorrect endpoint
- Network or connectivity problems

4.2 How to Use?

Sample Request

```
curl --location 'https://stg-apis.data.dubai/secure/ddads/healthcheck/1.0.0/health' \
--header 'Authorization: Bearer xxx
```

```
curl --location 'https://apis.data.dubai/secure/ddads/healthcheck/1.0.0/health' \
--header 'Authorization: Bearer xxx
```

Sample Response

```
{
   "status": "success",
   "statusCode": 200,
   "message": "API is healthy",
   "applicationName": "PUBLIC-USR-UID-123456"
}
```

Sample Request on Postman

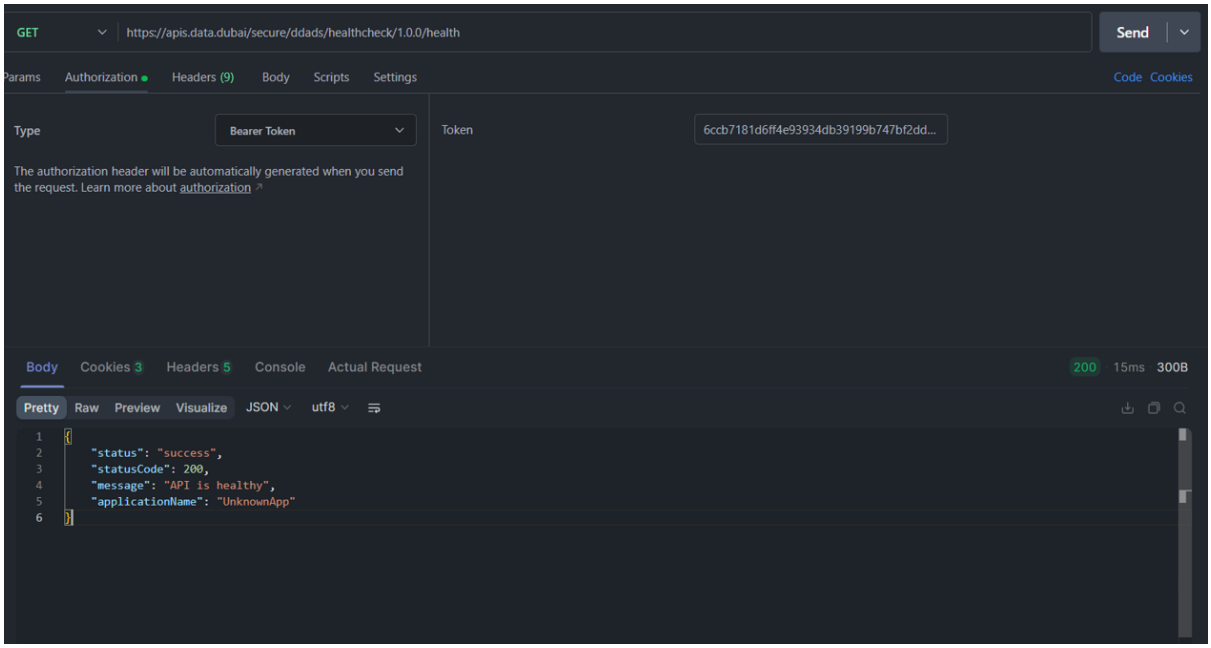


Figure: Health-Check API Postman Request

5. Invoking Data API

The Data API is the primary endpoint used to retrieve data from the data.dubai platform. Once authentication is completed and the Bearer Token is generated, this API can be invoked to access the required datasets.

5.1 API Details:

Key Details

- **Purpose:** Fetch data from data.dubai APIs
- **Authentication:** Requires a valid Bearer Token
- **Method:** GET

Request Requirements

/secure/ddads/openapi/1.0.0/

Endpoint Path

GET {{BASE_URL}}	{entity}/{dataset_name}

Path Parameters

Parameter	Description	Required
entity	Business entity identifier	Yes
dataset_name	Dataset name to be queried	Yes

Query Parameters

Parameter	Type	Description	Required

column	string	Comma-separated column names expected to retrieve in response	Optional
filter	string	Filter expression (filter=value) for filtering response based on any column value	Optional
page	integer	Page number (default: 1)	Optional
pageSize	integer	Records per page	Optional
limit	integer	Max records (alternative to pagination)	Optional
order_by	string	Column to sort by	Optional
order_dir	string	Global sort direction (asc / desc)	Optional
offset	integer	Record offset to get records after that offset.	Optional

Request Headers

Header Key	Description	Required
Authorization	Bearer token	Yes

Response

On successful execution, the API returns the requested data in a structured format, typically JSON. The response includes:

- Dataset records
- Status indicators

Each API response returns **1000 records per page**.

5.2 How to Use?

Sample Request

```
curl --location 'https://stg-apis.data.dubai/open/amaf/amaf_minor-open-
api?page=1&pageSize=100&order_by= billing_portion&order_dir=asc' \
--header 'Authorization: Bearer <access_token>'
```

```
curl --location 'https://apis.data.dubai/open/amaf/amaf_minor-open-
api?page=1&pageSize=100&order_by= billing_portion&order_dir=asc' \
--header 'Authorization: Bearer <access_token>'
```

Sample Response


```
{
  "results": [
    {
      "file_type": "OLNHZQPK93419700",
      "file_status": "VJEGRTXYYSJXVUWR2622792021478127",
      "file_creation_date": "2023-12-07",
      "minor_gender": "GFFHG9171",
      "load_timestamp": "2025-12-30 18:19:15"
    },
    {
      "file_type": "CQWDCDLO75159878",
      "file_status": "LHVHIYRIHJDLUQEI7438609349778299",
```

```
        "file_creation_date": "2025-03-28",
        "minor_gender": "EXZXL47946",
        "load_timestamp": "2025-12-30 18:19:15"
      },
      {
        "file_type": "RAGWMIGC99393364",
        "file_status": "VJVCIWLXYSNXKOAB4855389398263791",
        "file_creation_date": "2024-05-24",
        "minor_gender": "KZMSR59627",
        "load_timestamp": "2025-12-30 18:19:15"
      }
  ]
}
```


Sample Request on Postman

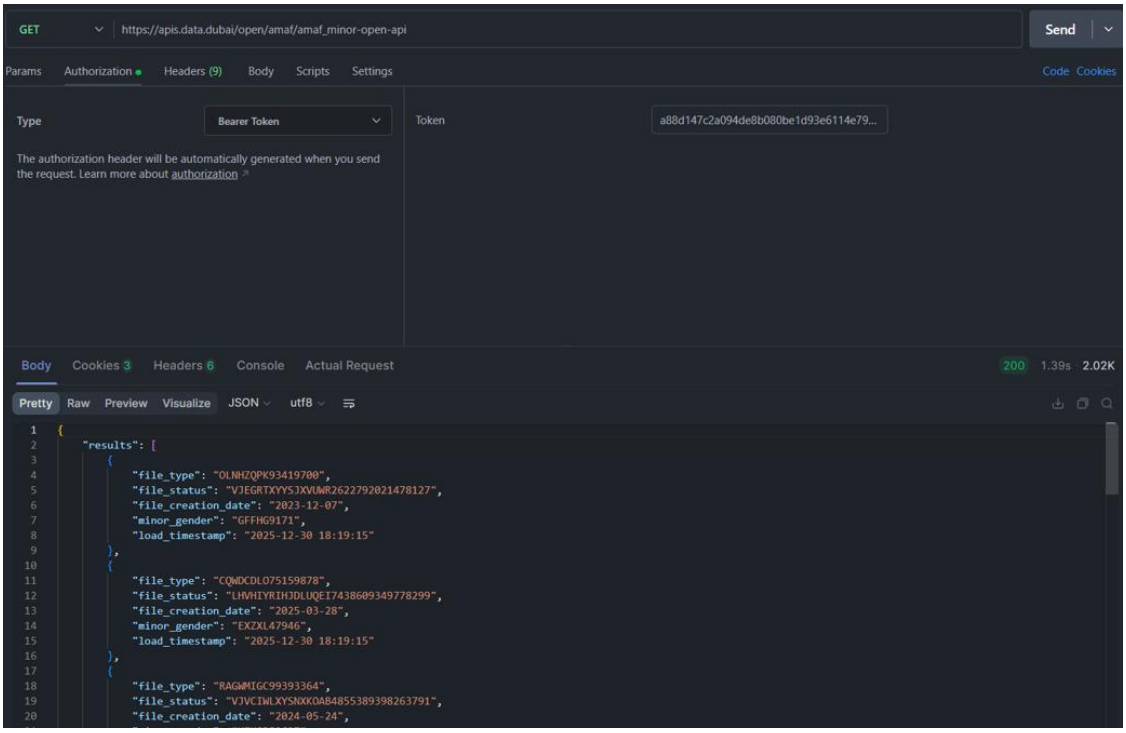


Figure: Data API Postman Request

In case of failure, the response will include appropriate error messages indicating the issue, such as invalid parameters, unauthorized access, or server errors.

6. Additional Information

Please take note of the following important considerations while using the data.dubai APIs:

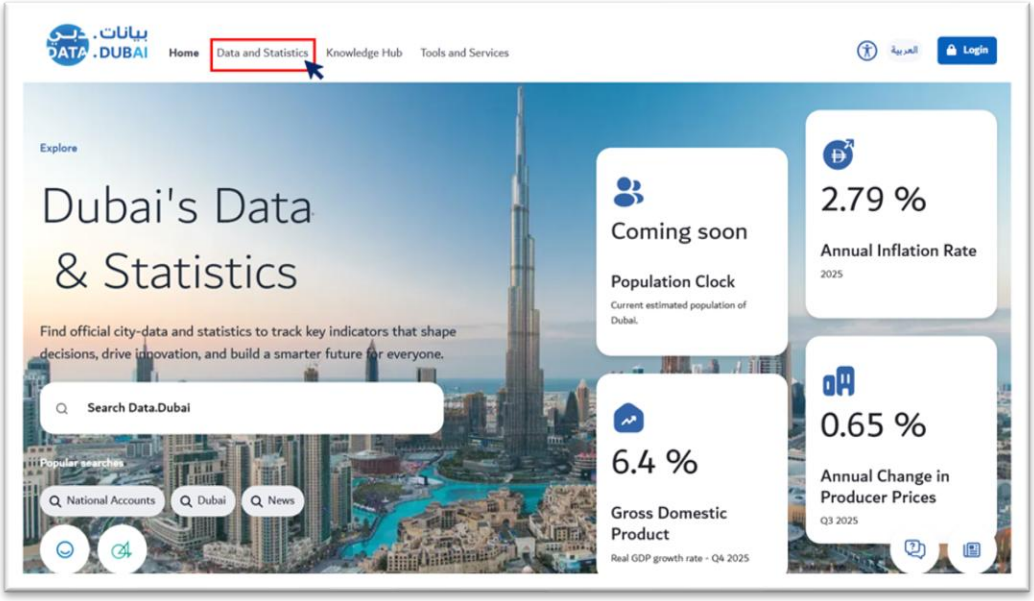
1. **API Rate Limit**
   The APIs are configured with a rate limit of **60 requests per minute**. Exceeding this limit may result in throttling or temporary request rejection.

2. **API Timeout**
   Each API request has a timeout threshold of **30 seconds**. If a response is not received within this duration, the request will be terminated.

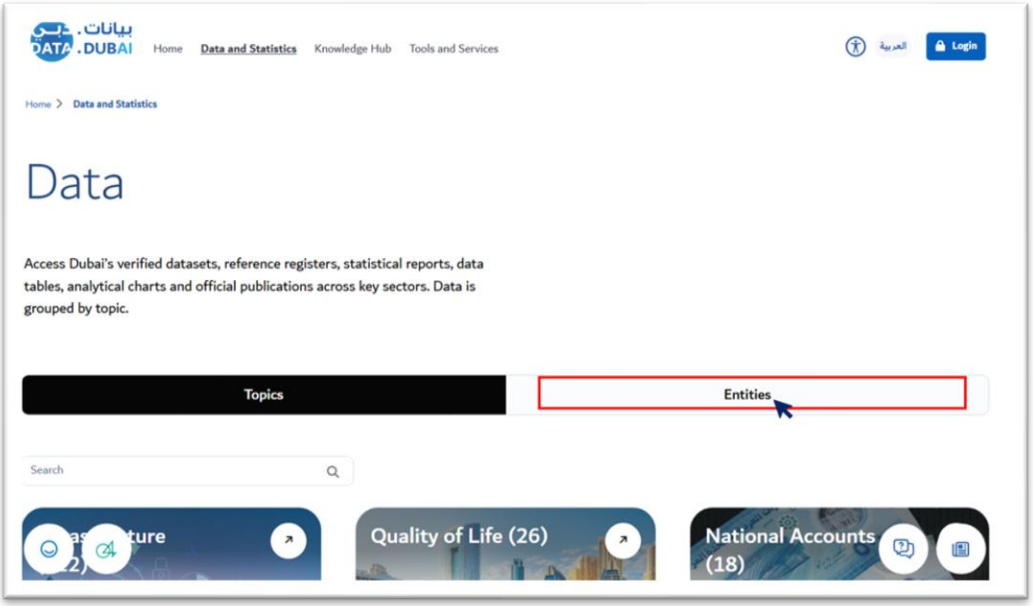
3. **API Accessibility**
   All the APIs can be accessed over the internet, **within the United Arab Emirates**. The APIs are inaccessible outside the country.

4. **API Credential Validity**
   The generated user credentials do not have any expiry, and any number of calls can be performed with the rate limit.

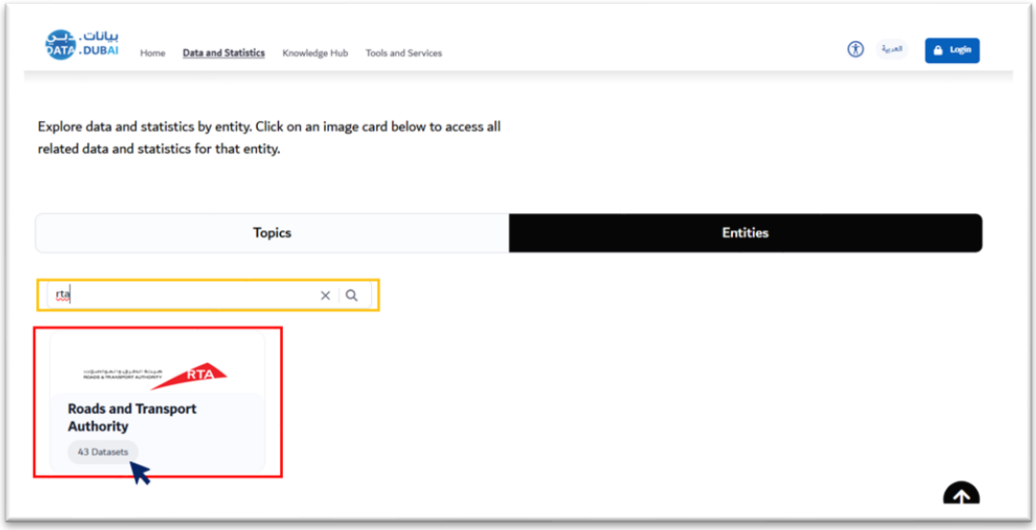
5. **Dataset Navigation**
   Step 1: Visit the data.dubai homepage and click on "**Data and Statistics**".



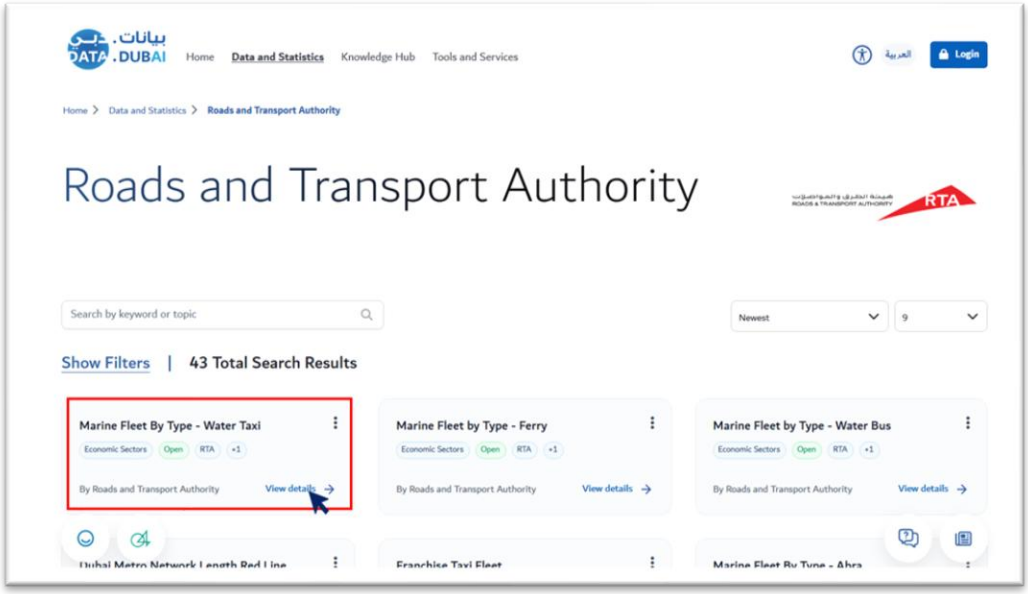
   Step 2: Now, on the "Data and Statistics" page, click on **Entities**. Choose from the available list of entities; if you are unable to find the desired entity, please try the next.



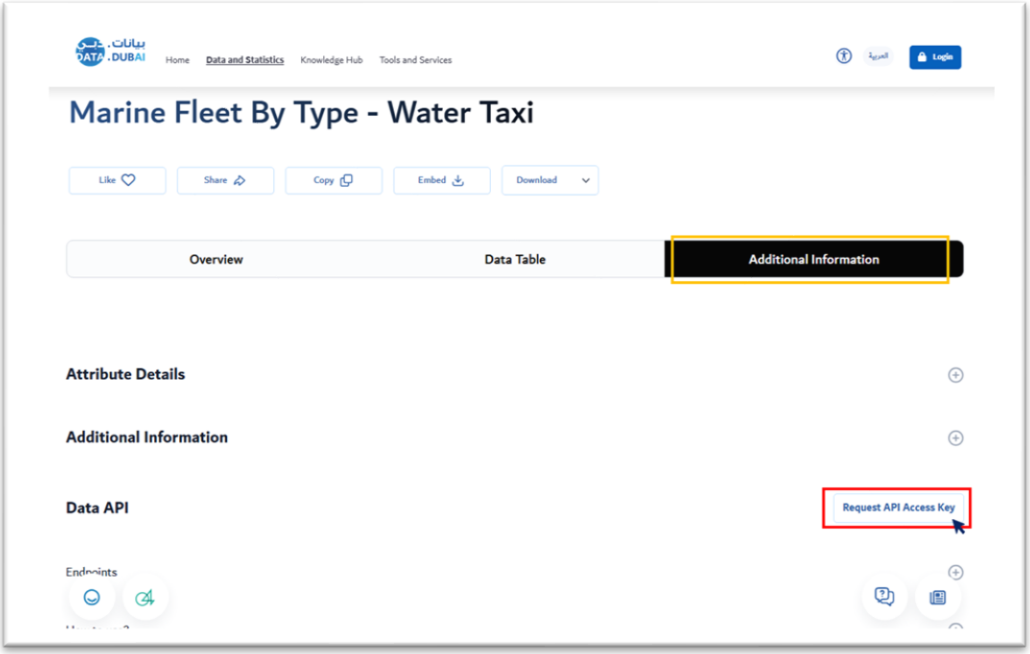
Step 3: To navigate to any desired entity, please use the search option as shown below.



Step 4: On the entity page, choose the desired dataset and click "**View details**".

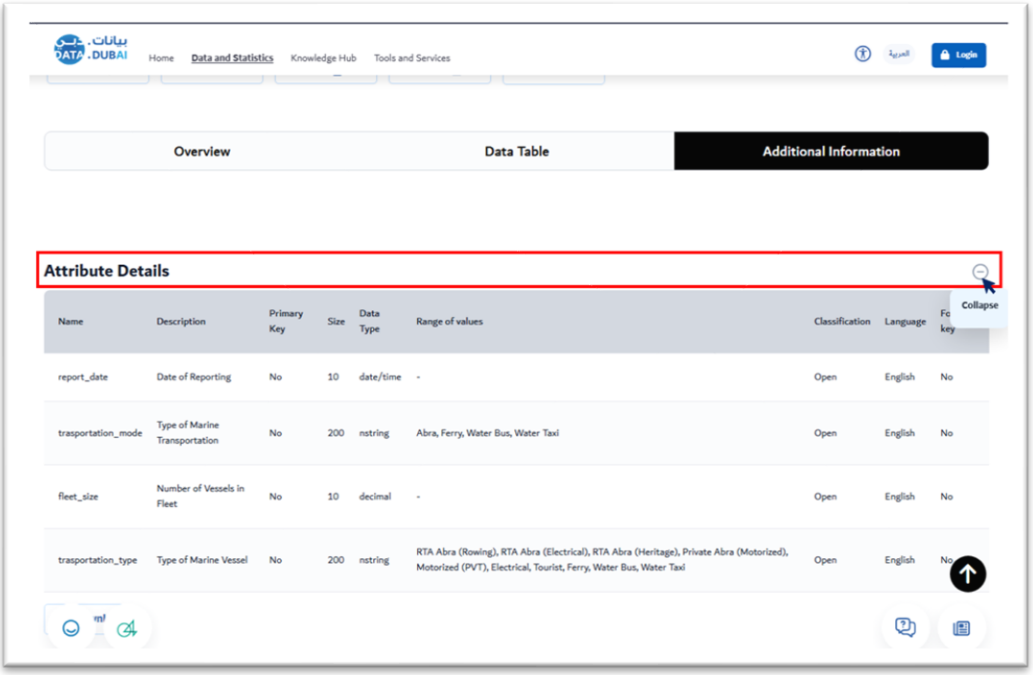


Step 5: On the dataset page, click on "**Additional Information**" tab. Under this section, navigate to the "**Request API Access Key**" to request access for the desired API. This is a one-time activity, and access will be granted as per requested classification.

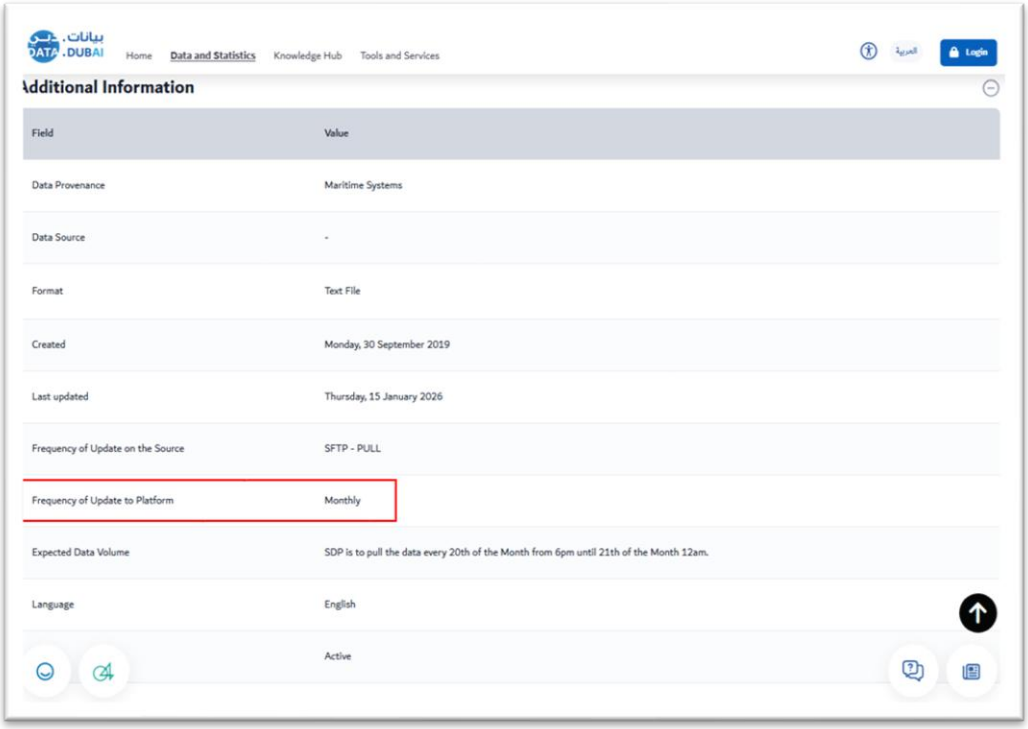


Note:

1. To learn more about the attribute details of a particular dataset, refer to the **'Additional information' – "Attribute Details"** section.



2. The dataset refresh frequency can also be viewed under "**Additional Information – Additional Information – Frequency of Update to Platform**" section



7. Support and Assistance

In case of any further questions, please submit your queries through the “Contact Us” form on the Data Dubai portal: https://data.dubai/en/contact-us